$$\frac{\partial^2 p}{(\partial x)^2} - \frac{1}{c^2}\frac{\partial^2 p}{(\partial t)^2} + \frac{2\gamma}{c^2}\frac{\partial^3 p}{\partial x^2 \partial t} = 0$$

$$p(x,t) = P_0 e^{-\alpha x}\cos(\omega t - kx)$$

$$\frac{\partial p}{\partial x} = P_0\left(-\alpha e^{-\alpha x}\cos(\omega t-kx) + e^{-\alpha x} k\sin(\omega t-kx)\right)$$

$$\frac{\partial^2 p}{\partial x^2} = P_0\left(\alpha^2 e^{-\alpha x}\cos(\omega t-kx) - \alpha e^{-\alpha x} k\sin(\omega t-kx) - \alpha e^{-\alpha x} k\sin(\omega t-kx) - e^{-\alpha x}k^2\cos(\omega t-kx)\right)$$
$$-2\alpha e^{-\alpha x} k\sin(\omega t-kx)$$

$$\frac{\partial p}{\partial t} = -P_0 e^{-\alpha x}\omega\sin(\omega t-kx)$$

$$\frac{\partial^2 p}{(\partial t)^2} = -P_0 e^{-\alpha x}\omega^2\cos(\omega t-kx)$$

$$\frac{\partial^3 p}{\partial x^2 \partial t} = P_0\left(-\alpha^2 e^{-\alpha x}\omega\sin(\omega t-kx) - 2\alpha k\omega e^{-\alpha x}\cos(\omega t-kx) + e^{-\alpha x}k^2\omega\sin(\omega t-kx)\right)$$

$\theta = \omega t - kx$

$$P_0\alpha^2 e^{-\alpha x}\cos(\omega t-kx) - 2P_0\alpha e^{-\alpha x} k\sin(\omega t-kx) - P_0 e^{-\alpha x}k^2\cos(\omega t-kx) + \frac{1}{c^2}P_0 e^{-\alpha x}\omega^2\cos(\omega t-kx) -$$
$$-\frac{2\gamma}{c^2}P_0\alpha^2 e^{-\alpha x}\omega\sin(\omega t-kx) - \frac{4\gamma}{c^2}P_0\alpha k\omega e^{-\alpha x}\cos\theta + \frac{2\gamma}{c^2}P_0 e^{-\alpha x}k^2\omega\sin\theta = 0$$

$$\cos\theta\left(\alpha^2 e^{-\alpha x} - k^2 e^{-\alpha x} + \frac{1}{c^2}e^{-\alpha x}\omega^2 - \frac{4\gamma}{c^2}\alpha k\omega e^{-\alpha x}\right) + \sin\theta\left(-2\alpha e^{-\alpha x}k - \frac{2\gamma}{c^2}\alpha^2 e^{-\alpha x}\omega + \frac{2\gamma}{c^2}k^2\omega e^{-\alpha x}\right) = 0$$

$$\begin{cases} \alpha^2 - k^2 + \dfrac{\omega^2}{c^2} - \dfrac{4\gamma}{c^2}\alpha k\omega = 0 \quad (1) \\ -2\alpha k - \dfrac{2\gamma}{c^2}\alpha^2\omega + \dfrac{2\gamma}{c^2}k^2\omega = 0 \quad (2) \end{cases}$$

$$\mu = \frac{\gamma\omega}{c^2}$$

(1): $\alpha^2 - k^2 + \frac{\omega^2}{c^2} - 2\mu\cdot 2\alpha k = 0$

(2): $2\alpha k = 2\mu(k^2 - \alpha^2)$ $\Rightarrow$

$\Rightarrow$ $\alpha^2 - k^2 + \frac{\omega^2}{c^2} - 4\mu^2(k^2-\alpha^2) = 0$

$(k^2-\alpha^2)(1+4\mu^2) = \frac{\omega^2}{c^2}$

$$k^2 - \alpha^2 = \frac{\omega^2}{c^2(1+4\mu^2)} \quad (3)$$

(2): $4\alpha^2 k^2 = 4\mu^2\frac{\omega^4}{c^4(1+4\mu^2)^2}$

$(k^2+\alpha^2)^2 = (k^2-\alpha^2)^2 + 4\alpha^2 k^2 =$

$$= \frac{\omega^4 + 4\mu^2\omega^4}{c^4(1+4\mu^2)^2} \quad (\text{из } 2 \text{ и } 3)$$

$$\begin{cases} k^2 + \alpha^2 = \dfrac{\omega^2}{c^2\sqrt{1+4\mu^2}} \\ k^2 - \alpha^2 = \dfrac{\omega^2}{c^2(1+4\mu^2)} \end{cases}$$

$$k^2 = \frac{\omega^2}{2c^2}\left(\frac{1}{\sqrt{1+4\mu^2}} + \frac{1}{1+4\mu^2}\right)$$

$$\alpha^2 = \frac{\omega^2}{2c^2}\left(\frac{1}{\sqrt{1+4\mu^2}} - \frac{1}{1+4\mu^2}\right)$$